

Richard Suhendra

📧: rsuhendra in: Richard Suhendra

✉: richard@suhendra.com

☎: (714)-722-3600

EDUCATION

- **Northwestern University** Evanston, IL
PhD in Applied Mathematics | GPA 3.89/4.0 2020 – 2026
- **The Pennsylvania State University** State College, PA
BS in Mathematics (Honors), Minor in Statistics | GPA 3.88/4.0 2016 – 2020

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Bash, SQL, R, MATLAB, Java
- **Machine Learning & AI:** scikit-learn, PyTorch, XGBoost, HuggingFace Transformers, LangChain, spaCy, statsmodels, OpenCV
- **Data/Visualization:** pandas, Snowflake, PySpark, Databricks, Redshift, PostgreSQL, Tableau
- **Cloud & Workflow:** AWS (EC2, S3, CLI), Docker, Metaflow, Airflow, dbt, Git
- **Selected Skills:** Machine Learning, Statistical Modeling, Large Language Models (LLMs), Multimodal AI, Fine-Tuning, Optimization, Causal Inference, High-Performance Computing

EXPERIENCE

- **Research Scientist Intern** 2025
Upstart (Machine Learning Department)
 - Built production-ready loan delinquency models on 500M+ records using neural networks and gradient-boosted trees for large-scale default risk prediction.
 - Developed a Markov chain-based forecasting model that chained XGBoost-estimated next-day delinquency probabilities to predict long-term default risk.
 - Improved default prediction over baseline models, decreasing calibration error by 13% and log loss by 5%; validated results through calibration metrics and leakage-aware backtesting.
- **Graduate Teaching Assistant** 2024 – 2026
Northwestern University
 - Designed and led hands-on NLP and machine learning labs, guiding implementations of transformers, RAG systems, and PyTorch-based deep learning workflows.
 - Revamped course infrastructure by migrating MATLAB assignments to Python, adding unit tests, and building a GitHub Actions auto-grader that automatically evaluated submissions.

RESEARCH PROJECTS

- **Mechanisms of adaptation to extreme temperatures in the snow fly** 2023 – 2025
Department of Neurobiology, Northwestern University
 - Built an end-to-end bioinformatics pipeline that transformed multimodal genomic data into protein sequences for downstream annotation and comparative analysis.
 - Fine-tuned Meta ESM2 protein LLM in PyTorch to identify antifreeze proteins ($F_1=0.80$), using model predictions to prioritize candidate proteins for biological interpretation.

- Analyzed RNA-seq data with DESeq2, PCA, and t-SNE to identify 50+ differentially expressed genes and validate cold-shock transcriptional response patterns.
- **Multimodal Music Recommendation System** 2026
Independent Machine Learning Project
 - Built a multimodal music recommender that converts images and text into music search briefs using Qwen2.5-VL, then retrieves matching tracks through CLAP text-audio embeddings.
 - Implemented context-aware query rewriting so follow-up user feedback updates CLAP search queries without discarding previous preferences.
- **A model for goal-directed navigation in a thermal gradient** 2023 – Present
Department of Engineering Sciences and Applied Mathematics, Northwestern University
 - Designed a mathematical model of temperature-guided navigation, linking behavioral data to candidate neural circuit mechanisms.
 - Engineered OpenCV-based computer vision pipelines to track fly movement trajectories and applied time-series analysis to uncover behavioral patterns.
 - Validated results using generalized linear mixed models (GLMMs) and nonparametric hypothesis testing to demonstrate significant differences in treatment.
- **Rapid threat assessment in the *Drosophila* thermosensory system** 2022 – 2023
Department of Engineering Sciences and Applied Mathematics, Northwestern University
 - Utilized Monte Carlo simulation to match stochastic agent-based models to fly movement patterns, integrating a neuronal derivative model to achieve a 20% reduction in overall error.
 - Developed multi-objective genetic algorithms in DEAP to optimize model parameters, utilizing a high-performance computing (HPC) environment to accelerate computations by 40x.
- **Optimizing robots with entropy-regularized Deep Reinforcement Learning** 2022
Deep Reinforcement Learning Final Project, Northwestern University
 - Utilized a Soft Actor-Critic (SAC) framework with deep neural networks in PyTorch to optimize OpenAI-gym robots, achieving a 120% increase in final average reward.
 - Implemented automatic hyperparameter tuning, further increasing convergence rate by 60%.

PUBLICATIONS

1. Capek M, **Suhendra R**, ... , Kath WL, Stensmyr M, Gallio M (2026). *Coordinated molecular and physiological adaptations enable activity at sub-freezing temperatures in the snow fly *Chionea alexandriana**. Current Biology.
2. Jouandet GC, Alpert MH, Simões JM, **Suhendra R**, ... , Kath WL, Gallio M (2023). *Rapid threat assessment in the *Drosophila* thermosensory system*. Nature Communications.
3. **Suhendra R** (2020). *Speeds of Biological Invasion*. Honors Thesis.
4. **Suhendra R**, Simões JM, ... , Kath WL, Gallio M. *A model for goal-directed navigation in a thermal gradient based on *Drosophila* behavior*. Expected submission: 2026